

Science · GROW · Autumn 1 · Week 08 · SCI-GROW-A1-W08-P2 · Date _____

Autumn Science: Show Your Thinking

Today I can... ☐ Explain one helpful and one unhelpful friction example ☐ Explain a mechanism gain and its cost ☐ Choose change, measure and controls for a fair comparison ☐ Explain an orbit and the Moon's reflected light

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____

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**Autumn Science: Show Your Thinking · Route card**

Reference only · record your answers in your book.

Independent

Answer the four decisions and support each with a science reason.

Supported

Use prepared cards for friction, a lever, a fair comparison and Moonlight.

Point to the evidence, then finish one because statement.

Choice

Answer all four decisions using the evidence cards. Choose: friction helps shoes grip / makes shoes float; friction at a rubbing axle helps / wastes energy. Choose the pivot: turning point / end of the lever; less effort force needs more / less effort distance. For a surface comparison, choose measure distance / colour and keep the block same / different. Choose gravity / reflected light keeps Earth in orbit; the Moon reflects sunlight / makes light. Finish one because reason for each decision. Record C beside your book response.

Word bank

- friction: a force opposing sliding or attempted sliding between touching surfaces
- pivot: the point a lever turns around
- control: something kept the same in a comparison
- orbit: a curved path around another body

Model steps

- Use genuine work if it is available; a missing record stays missing.
- A supplied comparison says the same block travelled 45 cm on smooth board and 19 cm on fabric. These are illustrative data, not your practical result.
- The evidence can support a claim about these surfaces and this block; one example is not every surface.

Kit: Genuine books, prepared evidence cards and familiar models.

Safe: Use prepared diagrams and cards. No new practical is required; handle any models under usual classroom routines.

Aspire

Explain each decision and qualify a model or data claim.

Identify what the evidence supports and one thing it cannot show.

Staff mark I, S or C beside this task.

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**Autumn Science: Show Your Thinking**

Reference only · record your answers in your book.

Learning goal: I can explain familiar science using an observation, model or genuine piece of work.

Word	Meaning
friction	a force opposing sliding or attempted sliding between touching surfaces
pivot	the point a lever turns around
control	something kept the same in a comparison
orbit	a curved path around another body

Key ideas

- Use genuine work if it is available; a missing record stays missing.
- A supplied comparison says the same block travelled 45 cm on smooth board and 19 cm on fabric. These are illustrative data, not your practical result.
- The evidence can support a claim about these surfaces and this block; one example is not every surface.
- A lever may reduce the effort force by increasing the effort distance. It does not create energy.

Keep the claim fair

Illustrative data support reasoning practice only. They are never recorded as a completed practical.

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**Staff: cut these apart before the lesson.**

Move these pieces for the task. Record the answer and reason in your book.

Brakes: Pads oppose sliding at the wheel.	Lever: The hand travels farther when less effort is needed.
Controls: Same block, release point and ramp; change the surface.	Moonlight: Sunlight reaches the Moon and reflects towards us.
Orbit: An inward pull changes a planet's direction.	Engine: Unwanted rubbing produces heat.

Method strip

1. Select the familiar evidence cards for the four science decisions.
 2. Identify which observation belongs to each decision. Stop and check the task and agreed support with the adult.
 3. Keep a first answer and use only the permitted support, with no new practical required.
 4. In your book, record the evidence, a scientific conclusion and one checked improvement.
- After the practical: record an observation and one conclusion linked to today's outcome in your book.